

VORTICES IN CURVED SPACE-TIME*

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We study an Abelian Higgs model coupled to a background metric. We find Bogomol'nyi equations when the coupling is achieved through an $R\phi^2$ term (R being the scalar curvature and ϕ the Higgs scalar). Remarkably, these equations coincide with those arising in models where the gauge field dynamics is governed by a Chern-Simons term so that vortex solutions in our system can be related to self-dual Chern-Simons vortices.

The analysis of the extended stable solutions to the classical nonlinear equations of motion of different field theories has considerably advanced after the observation that lower bounds for the energy (in Minkowski space) or the action (in Euclidean space) can be established in terms of topological charges.¹⁻³ These bounds, often called Bogomol'nyi bounds, are saturated by first order differential equations (Bogomol'nyi equations, self-duality equations) whose solutions solve the more involved second order Euler-Lagrange equations. In this way (multi) monopole and vortex solutions arising in gauge theories with spontaneous symmetry breaking were constructed and instanton solutions to Yang-Mills and CP^n models were discovered.¹⁻⁴

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More recently, Bogomol'nyi type lower bounds were found for relativistic and non-relativistic theories containing a Chern-Simons term and the corresponding topological and non-topological soliton solutions were constructed.⁵⁻⁷ Concerning quantum systems, Bogomol'nyi equations were shown to play a central role in the construction of topological quantum field theories (see Ref. 8 and references therein) and their connection with supersymmetry was stressed.⁹

The results described above correspond to flat space-time. Some of them have been extended so as to account for special background metrics¹⁰⁻¹⁵ and, in a few cases, the connection between Bogomol'nyi equations and Einstein equations has been analyzed. In particular, for vortices in general relativity, Linet¹⁴ and Comtet and Gibbons¹⁵ have studied solutions to an Abelian Higgs model with spontaneous symmetry breaking finding Bogomol'nyi equations which are compatible with Einstein equations. With respect to self-gravitating monopoles, the basic equations and properties of the solutions were written down by van Nieuwenhuizen, Perry and Wilkinson.¹² More recently Balakrishna and Wali¹⁶ have considered a Yang-Mills Higgs system coupled to gravity through an $R\phi^2$ term (R is the Ricci scalar and ϕ the Higgs field). They found Bogomol'nyi equations which are compatible with the gravitation equations whenever the Higgs field takes its vacuum expectation value everywhere in space while the metric can be interpreted as a generalization of the Majumdar-Papapetrou solution for the Einstein-Maxwell theory.¹⁷

Inspired by these results, we present in this work a study of the Abelian Higgs model in curved $(3+1)$ space-time showing that Bogomol'nyi equations can be found precisely when the Higgs scalar has an $R\phi^2$ coupling to the background metric. These equations saturate a lower bound (of topological character) for the energy of vortex configurations but, in contrast to the flat space case, there is no necessity of imposing the usual relation between the symmetry breaking coupling constant λ and the gauge coupling constant e ($\beta \equiv 8\lambda/e^2 = 1$) in order to have consistent equations.^{1,18} Instead, a relation between the metric and the Higgs scalar ϕ must hold, this leads to a decoupled equation for the scalar field. Surprisingly enough, this equation is the same as that arising in a $(2+1)$ -dimensional system with gauge field dynamics governed by a Chern-Simons action and a ϕ^6 term in the scalar potential.^{5,6} Our model then exhibits vortex solutions which can be related to those of the Chern-Simons Higgs system.

The action for the system we shall consider is

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} D_\mu \phi^a D^\mu \phi^a - V[\phi] - \frac{1-\beta}{2} B(x) \phi^2 \right), \quad (1)$$

where $g = \det g_{\mu\nu}$, ϕ^a ($a = 1, 2$) is a charged scalar, the covariant derivative D_μ is defined in terms of the gauge field A_μ as

$$D_\mu \phi^a = \partial_\mu \phi^a + e \varepsilon^{ab} A_\mu \phi^b \quad (2)$$

and $V[\phi]$ is the symmetry breaking potential:

$$V[\phi] = \frac{e^2}{8} \beta (\phi^2 - F^2)^2 \quad (3)$$

with

$$\beta = \frac{8\lambda}{e^2}. \quad (4)$$

Finally, $B(x)$ is a scalar source. Note that in curved space, when $\beta = 1$ the source term vanishes leading to the model which has been usually analyzed in the search of Bogomol'nyi equations.^{14,15}

The equations of motion arising from (1) are

$$\frac{1}{\sqrt{-g}} \partial_\mu \sqrt{-g} F^{\mu\nu} = -e\epsilon^{ab} \phi^a D^\nu \phi^b, \quad (5)$$

$$\frac{1}{\sqrt{-g}} D_\mu \sqrt{-g} D^\mu \phi^a = 4\lambda(\phi^2 - F^2)\phi^a + (1 - \beta)B\phi^a. \quad (6)$$

We are looking for Bogomol'nyi equations having vortex solutions and hence we shall consider a static cylindrically symmetric space-time. The metric will then be taken so that¹⁹:

$$ds^2 = V^2 dt^2 - Y^2 dz^2 - g_{ij} dx^i dx^j \quad (7)$$

with V , Y and g_{ij} ($i, j = 1, 2$) depending only on the polar coordinate r . Being two-dimensional, g_{ij} can be written in the form

$$g_{ij} = \Omega^2(r) \delta_{ij}. \quad (8)$$

The energy momentum tensor for the matter system with action (1) is

$$T_{\mu\nu} = -\frac{1}{2} F_\mu^\alpha F_{\nu\alpha} + D_\mu \phi^a D_\nu \phi^a - g_{\mu\nu} L. \quad (9)$$

From energy-momentum tensor conservation one can show that, for static spaces,¹³ the current:

$$j^\mu = \sqrt{-g} T_0^\mu \quad (10)$$

is conserved and the associated charge is precisely the energy for the configuration:

$$\mathcal{E} = \int d^3x \sqrt{-g} j^0. \quad (11)$$

Since we are looking for static, x^3 -independent (vortex-like) configurations we shall define an energy per unit length in the form:

$$E = \int d^2x \sqrt{-g} \left(\frac{1}{4} F^{ij} F_{ij} - \frac{1}{2} D^i \phi^a D_i \phi^a + V[\phi] + \frac{1-\beta}{2} B\phi^2 \right), \quad (12)$$

where as usual we have made $A_3 = 0$.

In order to find a bound for the energy, we make at this point the following ansatz:

$$Y^2 = \Omega^2 = V^{-2}. \quad (13)$$

Note that with this choice, the line element reduces to the isotropic form

$$ds^2 = V^2 dt^2 - \frac{1}{V^2} (dx^2 + dy^2 + dz^2) \quad (14)$$

adopted by Majumdar¹⁷ for finding a class of exact solutions to the Einstein-Maxwell equations. The same type of metric has led to *monopole solutions* in the work of Balakrishna and Wali¹⁶ and, as we shall see, also in the present work it leads to Bogomol'nyi equations for the Abelian Higgs model, with solutions of the vortex type. With (14) the energy as given by (12) can be written in the form:

$$\begin{aligned} E = \int d^2x \left\{ \frac{1}{4} \left[V F_{ij} \mp \frac{e\sqrt{\beta}}{2} \frac{1}{V} \varepsilon_{ij} (\phi^2 - F^2) \right]^2 \right. \\ + \frac{\beta}{4} V^{2(\beta-1)} \left[\varepsilon_{ij} V^{(1-\beta)} D_j \phi^a \mp \frac{1}{\sqrt{\beta}} \varepsilon_{ab} D_i (V^{1-\beta} \phi)^b \right]^2 \\ + \frac{\beta}{4} (1-\beta) V^{-2} \left[\varepsilon_{ab} \frac{1}{\sqrt{\beta}} D_i (\phi V)^b \pm \varepsilon_{ij} \phi^a \partial_j V \right]^2 \\ \left. + \frac{1-\beta}{4V^2} (2B - R) \phi^2 \pm F^2 \sqrt{\beta} \partial_i S_i \right\}. \end{aligned} \quad (15)$$

Given our choice of metric, the scalar curvature R takes the form:

$$R = 4(\partial V)^2 - 2V \partial^2 V. \quad (16)$$

The last term in (15) gives the surface contribution of the topological current S_i :

$$S_i = \frac{1}{2} \varepsilon_{ij} (\varepsilon_{ab} \hat{\phi}^a D_j \hat{\phi}^b + e A_j), \quad (17)$$

$$\oint S_i dx^i = \frac{2\pi}{e} n \quad (18)$$

with n the topological charge of the vortex configuration. (In Eq. (17) $\hat{\phi}^a = \phi^a/F$).

Let us note at this point that in the flat space limit, Eq. (15) can be easily reduced to the original Bogomol'nyi form (with the source $B = 0$). Indeed, making $V = 1$ one can rearrange it in the form:

$$\begin{aligned} E_{\text{flat}} = \int d^2x \left\{ \frac{1}{4} [F_{ij} \mp \varepsilon_{ij} (\phi^2 - F^2)]^2 + [\varepsilon_{ij} D_j \phi^a \mp \varepsilon_{ab} D_i \phi^b]^2 \right. \\ \left. + (\beta - 1) V [\phi] \pm \sqrt{\beta} \partial_i S_i \right\} \end{aligned} \quad (19)$$

and then a Bogomol'nyi bound for the energy can be found for $\beta = 1$.

Coming back to the curved space case, a lower bound for the energy can be found from Eq. (15) when the following conditions hold:

$$B = R/2 \quad (20)$$

and

$$\beta \leq 1. \quad (21)$$

Indeed, for this value, one has

$$E \geq \frac{2\pi\sqrt{\beta}}{e} |n| \quad (22)$$

and the bound is saturated whenever the following equations hold:

$$F_{ij} = \pm \frac{e\sqrt{\beta}}{2} \frac{1}{V^2} \varepsilon_{ij} (\phi^2 - F^2), \quad (23)$$

$$\varepsilon_{ij} V^{(1-\beta)} D_j \phi^a = \pm \frac{1}{\sqrt{\beta}} \varepsilon_{ab} D_i (V^{1-\beta} \phi)^b, \quad (24)$$

$$\varepsilon_{ab} \frac{1}{\sqrt{\beta}} D_i (\phi V)^b = \mp \varepsilon_{ij} \phi^a \partial_j V. \quad (25)$$

Note that Eq. (23) is just the usual Bogomol'nyi equation for F_{ij} in curved space-time.^{14,15} (Here the upper (lower) sign corresponds to positive (negative) vortex topological charge.) Concerning Eqs. (24) and (25), it is important to notice that they correspond to just one independent equation for the Higgs field. In fact, one can easily show that Eq. (25) can be obtained from Eq. (24) by just algebraic manipulations so that we shall take (23) and (24) as the independent pair of Bogomol'nyi equations. Moreover, the solutions to these equations satisfy the (second order) Euler-Lagrange Eqs. (7) and (8) *without imposing* $\beta = 1$, a necessary condition in all previous approaches to the construction of self-duality equations for vortex configurations.¹⁻¹⁸ In order to understand this peculiarity of our construction one has to note that for $\beta < 1$, Eq. (24) (or (25)) leads to a relation between the Higgs field and the metric. Indeed, starting from Eq. (25) and multiplying each side by $\varepsilon_{ac} \phi^c$ one obtains:

$$\partial_i \ln \phi^2 = -2\partial_i \ln V \quad (26)$$

and then, by adopting the usual asymptotic conditions $\lim_{r \rightarrow \infty} V^2 = 1$ and $\lim_{r \rightarrow \infty} \phi^2 = F^2$, one finds:

$$\phi^2 = F^2/V^2. \quad (27)$$

This result should be confronted with previous investigations in the flat space case and in curved space (but with the B -term absent). In these last cases, instead of Eq. (26) one gets, for $\beta \neq 1$, $\partial_i \phi^2 = 0$ or $\phi^2 = F^2$ everywhere and hence only $\beta = 1$ gives non-trivial solutions. In a sense, Eq. (27) shows that in our construction, due

to the $R\phi^2$ coupling, the scalar field ϕ is traded for V , this allowing for non-trivial vortex solutions, as it happens for monopoles in an analogous model.¹⁶

Concerning the condition (20), let us note that it corresponds to taking the external source equals to the scalar curvature interacting with the Higgs field through a coupling which has already been considered in several models of matter coupled to gravitation.¹⁶ We shall now use Eqs. (24) and (27) in order to decouple the scalar field from A_μ . To this end, we define

$$\phi = Fg \exp(i\alpha) \quad (28)$$

with g and α two real functions. Then, from Eq. (24) the gauge field can be written in the form:

$$eA_i = \partial_i \alpha \mp \frac{\sqrt{\beta}}{2} \epsilon_{ij} \partial_j \ln g^2. \quad (29)$$

Inserting (29) into (23) we obtain

$$\partial^2 \ln g^2 = \frac{e^2 F^2}{V^2} (g^2 - 1) \quad (30)$$

and, after (27), we have

$$\partial^2 \ln g^2 = e^2 F^2 g^2 (g^2 - 1). \quad (31)$$

Remarkably, this is precisely the same equation obeyed by the Higgs field in the Abelian Chern-Simons theory with spontaneously symmetry breaking found by Jackiw and Weinberg⁶ (see also Ref. 5). Moreover, Eq. (23) together with Eq. (27) reduces to the equation for the magnetic field of self-dual Chern-Simons vortices:

$$\frac{e}{2} \epsilon_{ij} F_{ij} = \pm \frac{m_V^2 \sqrt{\beta}}{2} g^2 (g^2 - 1). \quad (32)$$

(Here $m_V = eF$.) Equations (31) and (32) show that the vortex solutions for our system must be identical to those corresponding to self-dual Chern-Simons vortices.^{5,6} Nevertheless, it is instructive to directly analyze Bogomol'nyi equations (23)–(25) to see the interplay between the Higgs field and the metric. Let us make the usual ansatz for an n -vortex configuration:

$$\phi^1 = Fg(r) \cos n\varphi, \quad (33)$$

$$\phi^2 = Fg(r) \sin n\varphi, \quad (34)$$

$$eA_\varphi = -\frac{A(r) - n}{r}. \quad (35)$$

We then have for the metric (Eq. (27)):

$$V^2 = g^{-2}(r) \quad (36)$$

and, from Eqs. (24) and (25)

$$rg'(r) = \mp \frac{A(r)}{\sqrt{\beta}} g(r). \quad (37)$$

Concerning Eq. (23), one gets

$$\frac{A'(r)}{\sqrt{\beta}} = \mp \frac{m_V^2}{2} rg^2(g^2 - 1). \quad (38)$$

Calling $A(r)/\sqrt{\beta} = a(r)$, we then have

$$-g'(r) = \pm \frac{a(r)g(r)}{r}, \quad (39)$$

$$-\frac{a'(r)}{r} = \pm m_V^2 g^2(g^2 - 1) \quad (40)$$

which coincide with the Bogomol'nyi equations for the self-dual Chern-Simons system, thus having the same solutions as those described by Jackiw and Weinberg.⁶ For definiteness, consider the vortex solution with $n > 0$, this corresponding to the lower sign in Eqs. (39) and (40). Then, the vortex has the following behavior at infinity:

$$g(r) \approx 1 - cK_0(m_V r) \quad a(r) \approx cm_V r K_1(m_V r). \quad (41)$$

Concerning the origin, the field behavior can be determined by making a power series expansion.^{6,18} In particular, one finds that $g \sim r^{|n|}$ for small r which corresponds to a true singularity of the metric as can be seen from the behavior of the Ricci scalar at $r = 0$ (Eq. (16)). A similar situation occurs in the monopole case¹⁶; its connection with black hole physics deserves a thorough investigation.

An interesting feature of our vortex solution is that, as it happens in the case of self-dual Chern-Simons system, the corresponding magnetic field is concentrated in a cylindrical shell around the (n th order) zero of the Higgs field (while in the usual Abelian Higgs model it is the greatest at the center of the vortex). It should be stressed however that our solution and that of the Chern-Simons-Higgs model are physically distinct, this as a consequence of the fact that they correspond to space-times with different dimensions. In the $2 + 1$ Chern-Simons system vortices are both electrically and magnetically charged (in fact, the presence of the Chern-Simons term forces the solution to have this property). Ours, are static, axially symmetric solutions in $3 + 1$ space-time with $A_0 = 0$ and hence the electric field is zero. We can nevertheless understand the connection between both systems by the following analysis. Being the Chern-Simons term metric-independent, it does not enter in the corresponding energy-momentum tensor. Now, the gauge field dynamics in the Chern-Simons-Higgs system is solely controlled by this term and the Higgs field covariant derivatives. It is then through the latter that the gauge field enters in $T_{\mu\nu}$. In particular, the time component of this derivative gives an $A_0^2 \phi^2$

contribution to T_{00} . Using the time component of the static Chern-Simons-Higgs equations of motion, the A_0 dependence can be eliminated in terms of the magnetic field $\varepsilon_{ij} F_{ij}$ this leading to a $F_{ij}^2 \phi^{-2}$ term in the energy. But this is precisely the kind of contribution arising from the Maxwell term $\sqrt{-g} F_{ij}^2$ for T_{00} in our model, once the relation between g and ϕ^2 is used. Concerning the Higgs potential, it has a ϕ^6 term in the self-dual Chern-Simons-Higgs system which corresponds to the ϕ^4 potential for our model once the connection between g and ϕ^2 is again used (one, of course, has to take into account the $R\phi^2$ term in order for both equations to match).

Let us end our work by summarizing its main goals. Starting from the Abelian Higgs model coupled to a background metric we have shown that if this coupling is achieved through an $R\phi^2$ term, a Bogomol'nyi bound can be derived provided that an adequate (isotropic) ansatz for the metric is proposed. The self-duality equations arising from saturation of this bound coincide with those obtained for the self-dual Chern-Simons-Higgs system. Our construction holds for $\beta = 8\lambda/e^2 < 1$ and implies a definite relation between ϕ^2 and $\det g_{\mu\nu}$. Concerning the condition $\beta < 1$, it should be contrasted with the flat-space time result^{1,18} (as well as with the previous investigations in curved space-time^{14,15}) for which $\beta = 1$ was a necessary condition in order to find first order self-duality equations. Let us finally mention that vortex solutions can be simply analyzed from the Bogomol'nyi equations we present: they correspond to a magnetic field concentrated in a cylindrical shell around the zero of the Higgs field.

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